IIT Guwahati - Department of Computer Science & Engineering

CS 2201 – Computer Architecture & Organization

Quiz #2 - (15.10.2026: 90 minutes, 40 marks)

Index the facing sheet with question numbers and corresponding page numbers.

Q. No	Q:1	Q:2	Q:3	Q:4	Q:5	Q:6	Q:7	Q:8	Q:9	Q:10	Total
Max. Marks	3	3	2	2	3	3	4	6	6	8	40

1. Answer the following.
   (a) Name the lab that designs and fabricates processors for ISRO satellites.
   (b) Name the RISC processor brand developed by CDAC.
   (c) Name the most powerful supercomputer located in IIT Guwahati.
   (d) Who is the current CEO of Intel?
   (e) Who is the current chairman of ISRO?
   (f) Which Indian state houses Tata Electronics-PSMC fab facility?

   Ans: **0.5 marks for each correct sub-question**

   (a)Semi-Conductor Laboratory (SCL), Mohali

   (b)VEGA

   (c)PARAM Kamrupa

   (d)Lip-Bu Tan

   (e)Dr. V. Narayanan

   (f)Gujarat

2. Match the following with unique mapping from Set X to Set Y.

Set X	Set Y
(a) Way prediction	1. exploits sequential interleaving
(b) Non-blocking cache	2. allows faster clock cycle time
(c) Pipelined cache	3. implementation needs MSHR
(d) Multi-banked cache	4. reduces hit time
(e) Hardware pre-fetching	5. reduces miss penalty
(f) Victim cache	6. implementation needs stream buffer

   Ans: **0.5 marks for each correct match**

   a) 4

   b) 3

   c) 2

   d) 1

   e) 6

   f) 5

3. For a cache, that has a miss rate of 3% and miss penalty of 10 cycles, find the MPKI, if 50% of the instructions are memory accesses.

   Memory Accesses Per Instruction (MAPI) = 1 (instruction fetch) + 0.50 (memory access instructions)

   MAPI = 1.50

   Misses Per Kilo Instructions (MPKI) = Miss Rate x MAPI x 1000 [1 mark]

   MPKI = 0.03 x 1.50 x 1000

   MPKI = 45 [1 mark]

4. A cache has a hit time of 10 ns and miss rate is 5%. An optimization was made to reduce the miss rate to 3% but the hit time was increased to 15 ns. What should be the miss penalty if this change must result in reduction in average memory access time?

   Let the miss penalty be P ns.

Parameter	Original Cache	Optimized Cache
Hit time	10 ns	15 ns
Miss rate	5%	3%
Miss penalty	P ns	P ns

   AMAT = Hit time + Miss rate x Miss penalty

   $AMAT_{original} = 10 + 0.05P$

   $AMAT_{optimized} = 15 + 0.03P$

   $AMAT_{optimized} < AMAT_{original}$ [1 mark]

   $15 + 0.03P < 10 + 0.05P$

   $5 < 0.02P$

   $P > 250$ ns [1 mark]

   To achieve the reduction in AMAT, the miss penalty must exceed 250 ns.

5. A cache has hit rate of 95%, 128-byte block, and hit time of 5ns. Main memory takes 100 ns to return first word (16 bits) of a block and 10 ns for each subsequent word. If doubling the cache block size reduces the miss rate to 3%, does it reduce average memory access time? Why/why not?

   Hit rate = 95%
   Block size = 128B
   Hit time = 5ns
   1 word takes 100 ns
   word size = 16 bits = 2 B
   10ns for subsequent words

No of words = 128/2 = 64
Miss penalty = Time taken by 1st word + Subsequent 63 words
= 100 + 63*10
= 730 [0.5 mark]
AMAT = Hit rate + Miss rate * Miss penalty
= 5 + 0.05*730
= 41.5 [1 mark]
Doubling cache block size,
New block size = 128*2 = 256
No. of words = 128
Miss penalty = 100 + 127*10
= 1370 [0.5 mark]

AMAT = 5 + 0.03*1370
= 46.1 [1 mark]
No, doubling the cache block size increases AMAT.

6. Consider a processor with 0.5 ns clock rate and has two levels of cache. The cache hit takes 4 clock cycles for L1 cache. The miss rate for the L1 cache is 10%, and L2 cache is 5%. The miss penalty of L2 cache is 150 cycles. What is the hit time (in clock cycles) of L2 cache is if the average memory access time for this processor is 5.5 ns?

Clock rate = 0.5ns
L1 hit time=4 cycles
Miss rate of L1 = 10%
Miss rate of L2 = 5%
Miss penalty = 150
AMAT = 5.5

AMAT in cycles = 5.5/0.5
= 11 cycles [1 mark]

(Or convert all in ns and solve and write the final solution in clock cycles)

AMAT = Hit time of L1 + Miss rate of L1 (Hit time of L2 + Miss rate L2 * Miss penalty of L2) [1 mark]

11 = 4 + 0.1 (Hit rate of L2 + 0.05 * 150)

Hit time = 62.5 cycles [1 mark]

7. The address of a word in a byte addressable 1 MB physical memory is 0xAA0F2. This word upon bringing to a directly mapped cache is mapped to set 30. The word length of the processor is 16 bits.
(a) How many words can be accommodated in each cache block?
(b) What is the maximum and minimum capacity of cache under which the above scenario can happen?

Given

Main Memory =1 MB =2^{20} B —>20
1 word =16 bits=2B
Direct Mapping — > 1 way
0xAA0F2 = 1010 1010 0000 1111 0010

Set 30=11110 , so set index needs at least 5

(a) Words per block **[Total 1 mark]**

1010 1010 0000	11110	010
——————12——————	——————5——————	——————3——————
Tag	Set	Block Offset

Block offset =3 bits [0.5 Partial Marking]

Block size = 2^3 =8 B

Words per block = 8/2=4 [0.5 Partial Marking]

(b) Maximum and Minimum Capacity of Cache

12	**5**	**3**
Tag	Set	Block Offset

Minimum : #set =Cache Size/(Block Size * Set Associvity) [Total 1.5 marks]

$2^5 = CS / (2^3 * 2^0)$

$CS = 2^5 * 2^3 = 2^8$

CS= 256 B

Maximum: #set =Cache Size/(Block Size * Set Associvity) [Total 1.5 marks]

Set index can be 10 bits long so #sets= 2^{10} =1024

Block size = 2^3 =8 B

$2^{10} = CS / (2^3 * 2^0)$

$CS = 2^{10} * 2^3 = 2^{13}$ = 8192B= 8KB

8. Consider a processor Quantum with two variants Alpha and Beta. Alpha configuration uses I-cache and D-cache of 16 KB each. The I- cache has MPKI of 4 and D- cache has 42. Beta uses a unified 32KB cache with MPKI = 44. Assume 32 % instructions are load store instructions, and cache hit takes 1 clock cycle and miss penalty is 120 clock cycles. If the unified cache has only one port and hence loads and store takes one more clock cycle extra to satisfy two simultaneous cache requests, which variant of Quantum has lesser memory access time? Explain with necessary calculations.

Given

Alpha Processor:

I-Cache = 16 KB => 4 MPKI

D-Cache = 16 KB => 42 MPKI

Beta Processor = 32 KB => 44 MPKI

$$\text{Miss Rate} = \frac{Misses}{Memory\ Access}$$

Miss Rate = $\frac{MPKI\ /\ 1000}{\frac{Memory\ Access}{Instructions}}$

For 100 instructions, $\frac{Memory\ Access}{Instructions}$

of instruction access = 100 => 1

of data access = 32 => 0.32

Total number of access = 132 => 1.32

% share of instruction fetch out of total access = 100/132 = 76%

% share of data fetch out of total access = 32/132 = 24%

$\text{Miss Rate}_{\text{16 KB I Cache}} = \frac{4/1000}{1} = 0.004$

$\text{Miss Rate}_{\text{16 KB D Cache}} = \frac{42/1000}{0.32} = 0.13125$

Total Miss Rate of Alpha Processor = 76% * 0.004 + 24% * 0.13125 = 0.0345

$\text{Miss Rate}_{\text{32 KB Beta}} = \frac{44/1000}{1.32} = 0.0333$

[1 Mark for Miss Rate of Alpha Processor => if you don't write the % share of instruction fetch out of total access and % share of data fetch out of total access => 0 Marks]

[1 Mark for Miss Rate of Beta Processor]

Beta processor has lower miss rate than Alpha processor

AMAT = % instruction (Hit Time + Miss Rate x Miss Penalty) +

% data (Hit Time + Miss Rate x Miss Penalty)

$\text{AMAT}_{\text{Alpha}}$ = 76% (1 + 0.04 * 120) + 24% (1 + 0.13125 * 120)

= 1.1248 + 4.02 = 5.144

[2 Marks for AMAT alpha processor if you don't write the % share of instruction fetch out of total access and % share of data fetch out of total access => 0 Marks]

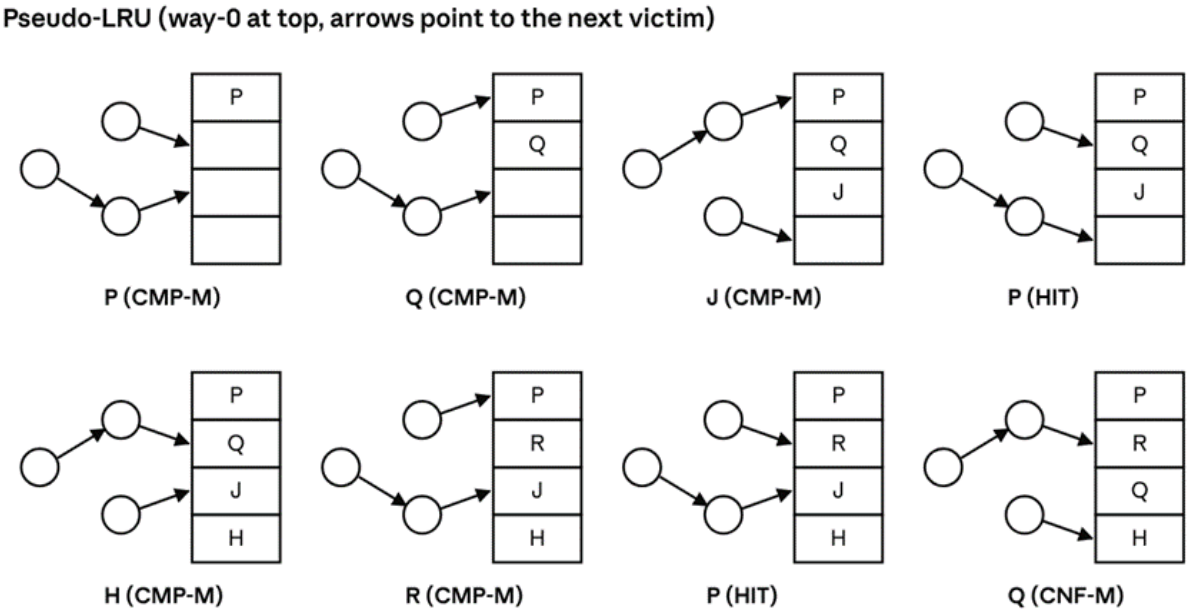
$\text{AMAT}_{\text{Beta}}$ = 76% (1 + 0.033 * 120) + 24% (1 +1 + 0.033 * 120)

= 3.76 + 1.549 = 5.309

[2 Marks for AMAT beta processor=> 1 Mark for extra stall]

9. The following 8 memory block requests P, Q, J, P, H, R, P, and Q are mapped to set n of a 4-way set-associative cache memory that uses two replacement techniques; LRU and Pseudo LRU. Assume that set n is initially empty. Assume that data is entered into an empty cache block by filling from way-0, way-1, way-2 & way-3 order. For each of the cache replacement techniques, compute the following with necessary illustrations.
(a) Show the PLRU tree and LRU steps after processing each of the above block requests.
(b) How many hits, compulsory misses and conflict misses are there from the above 8 requests?
(c) What will be the contents of set n (in the order way-0, way-1, way-2 and way-3) after servicing above 8 requests?

(a) Requests to set n: P, Q, J, P, H, R, P, Q
Cache: 4-way set associative, set n initially empty.



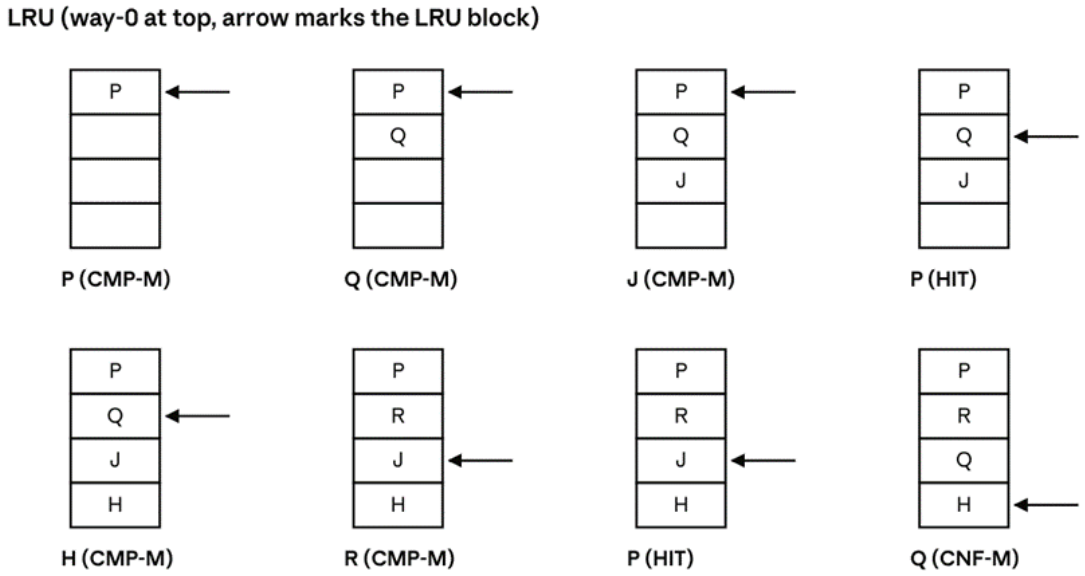
PLRU figure [2 mark]

Recency order is written from MRU (most recently used) to LRU (least recently used).

Step	Request	Result	Recency (MRU to LRU)	Way-0	Way-1	Way-2	Way-3
1	P	CMP-M	P	P	-	-	-
2	Q	CMP-M	Q, P	P	Q	-	-

3	J	CMP-M	J, Q, P	P	Q	J	-
4	P	HIT	P, J, Q	P	Q	J	-
5	H	CMP-M	H, P, J, Q	P	Q	J	H
6	R	CMP-M (Q replaced)	R, H, P, J	P	R	J	H
7	P	HIT	P, R, H, J	P	R	J	H
8	Q	CNF-M (J replaced)	Q, P, R, H	P	R	Q	H

Note: the recency order is not the way order. A new block takes the way of the block it replaces.



LRU figure [2 mark]

(b) Hits and misses [1/2 mark each for both]

Policy	Hits	Compulsory misses	Conflict misses
LRU	2	5	1
PLRU	2	5	1

(c) Final contents of set n [1/2 mark each for both]

Policy	Way-0	Way-1	Way-2	Way-3
LRU	P	R	Q	H

PLRU	P	R	Q	H

10. Consider a computer system having a processor that uses 16-bit instruction word and 16-bit data word. The processor is connected to an I-cache and D-cache of the same configuration. Both the caches are of 256 Bytes and use direct mapping with a block size of 16 Bytes. The system has a physical memory of 16 MB. A program consisting of 10 instruction is stored in memory from location 0x12820E. The program has a loop that adds two arrays B[16] and C[16] and store the result in A[16]. Each element in the array is 16 bit. The loop is iterated from index 0 to 15. The physical addresses of A, B, and C start from 0x432536, 0x502350, and 0xB32AB6, respectively. Assume that both the caches are initially empty. A, B, and C are the only data items accessed by the program.
    (a) Identify the tag, set-index and block-offset split-up of the physical address.
    (b) Find the number of read and write hits in D-cache.
    (c) Find miss rate of D-cache.
    (d) List the non-empty sets of both I-cache and D-cache after the program execution.
    (e) Which all elements of arrays A, B, and C are in the D-cache after the program execution.

Ans:
(a)
Block offset = 4 bits
set-Index = 4 bits
Tag =16 bits

Tag	Index	Block Offset
16	4	4

[1mark (No Partial Marking)]

B (Address)	**Set Index**	**Hit/Miss**	**B (Address)**	**Set Index**	**Hit/Miss**
0x502350	5	M	0x502360	6	M
0x502352	5	H	0x502362	6	H
0x502354	5	H	0x502364	6	H
0x502356	5	H	0x502366	6	H
0x502358	5	H	0x502368	6	H
0x50235A	5	H	0x50236A	6	H
0x50235C	5	H	0x50236C	6	H
0x50235E	5	H	0x50236E	6	H

C (Address)	Set Index	Hit/Miss	C (Address)	Set Index	Hit/Miss
0xB32AB6	B	M	0xB32AC6	C	H
0xB32AB8	B	H	0xB32AC8	C	H
0xB32ABA	B	H	0xB32ACA	C	H
0xB32ABC	B	H	0xB32ACC	C	H
0xB32ABE	B	H	0xB32ACE	C	H
0xB32AC0	C	M	0xB32AD0	D	M
0xB32AC2	C	H	0xB32AD2	D	H
0xB32AC4	C	H	0xB32AD4	D	H

A (Address)	Set Index	Hit/Miss	A (Address)	Set Index	Hit/Miss
0x432536	3	M	0x432546	4	H
0x432538	3	H	0x432548	4	H
0x43253A	3	H	0x43254A	4	H
0x43253C	3	H	0x43254C	4	H
0x43253E	3	H	0x43254E	4	H
0x432540	4	M	0x432550	5	M
0x432542	4	H	0x432552	5	H
0x432544	4	H	0x432554	5	H

(b) D-cache

Read hits= 27, **[1 mark]**

Write hits=13, **[1 mark]**

Total=40

(Tables or similar representation should be there (No marks if no description/explanation is given, even if the final answers are correct)

(c) Total misses= 8

Total access = 48

Miss Rate = 8/48 * 100 = 16.67 % **[1mark (No Partial Marking)]**

(d)

Program	Set Index	Hit/Miss

Instruction		
0x12820E	0	M
0x128210	1	M
0x128212	1	H
0x128214	1	H
0x128216	1	H
0x128218	1	H
0x12821A	1	H
0x12821C	1	H
0x12821E	1	H
0x128220	2	M

Now, non empty sets,

I-Cache: set 0,1,2 **[1mark]**

D-Cache: set 3,4,5,6,B/11,C/12,D/13 **[1mark]**

(Tables or similar representation should be there (No marks if no description/explanation is given, even if the final answers are correct))

(e) Elements of A, B & C in D-Cache after execution,

B= 8-15 ; A will evict B (set 5; 0-7)

C= 0-15

A= 0-15

[2 marks (description/explanation should be there for full marks)]